

STUDENT NUMBER/NAME:

GOSFORD HIGH SCHOOL

2014

PRELIMINARY COURSE

FINAL EXAMINATION

Mathematics Extension 1

- **General Instructions**

- Reading time – 5 minutes
- Working time – 1½ hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- In Questions 6-9, show relevant mathematical reasoning and/or calculations

Total Marks – 53

Section I

5 marks

- Attempt Questions 1 – 5
- Allow about 10 minutes for this section

Section II

48 marks

- Attempt Questions 6 – 9
- Allow about 1 hour and 20 minutes for this section

Section I

5 marks

Attempt Questions 1 – 5.

Allow about 10 minutes for this section.

Use the separate multiple-choice answer sheet for Questions 1 – 5.

1. Which of the following is an expression for $\frac{9^n - 6^n}{9^n - 4^n}$?

(A) $\frac{6^n}{4^n}$

(B) $\frac{3^n}{5^n}$

(C) $\frac{3^n}{3^n - 2^n}$

(D) $\frac{3^n}{3^n + 2^n}$

2. If $R\cos(x + \alpha) \equiv \sin x - \cos x$, what are the values of $R\cos\alpha$ and $R\sin\alpha$?

(A) $R\cos\alpha = -1$, $R\sin\alpha = -1$

(B) $R\cos\alpha = -1$, $R\sin\alpha = 1$

(C) $R\cos\alpha = 1$, $R\sin\alpha = -1$

(D) $R\cos\alpha = 1$, $R\sin\alpha = 1$

3. What is the value of $\frac{{}^nC_3}{{}^nP_3}$?

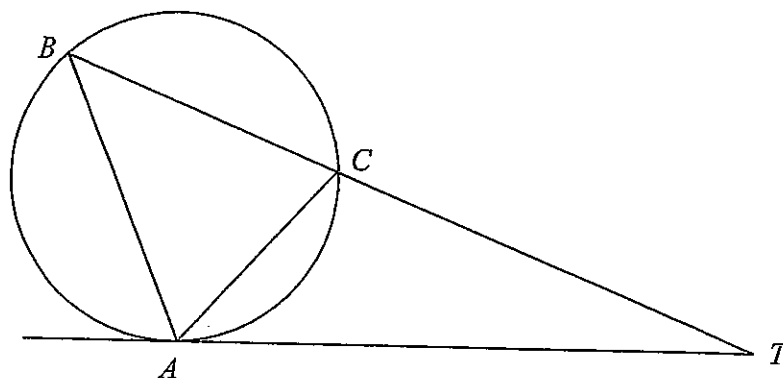
- (A) 6
- (B) 3
- (C) $\frac{1}{3}$
- (D) $\frac{1}{6}$

4. If $7\sqrt{x^5} - \sqrt{4x} + 3\sqrt{x^6} = a + b\sqrt{x}$, find a and b .

- (A) $a = 3x^3$, $b = 7x^2 - 4$
- (B) $a = 3x^3$, $b = 7x^2 - 2$
- (C) $a = 3x^2$, $b = 7x^2 - 2$
- (D) $a = 3x^6$, $b = 7x^4 - 4$

5.

NOT TO SCALE



In the diagram, ABC is a triangle inscribed in a circle. The tangent to the circle at A meets BC produced at T . $TA = 10$ cm and $TC = 8$ cm. What is the length of BC ?

- (A) 3.5 cm
- (B) 4 cm
- (C) 4.5 cm
- (D) 5 cm

END OF SECTION I

SECTION II

48 marks

Attempt Questions 6 – 9

Allow about 1 hour and 20 minutes for this section.

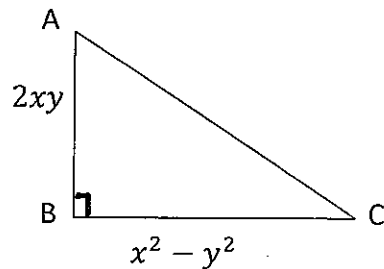
Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 6 – 9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6: (12 marks) (Use a SEPARATE writing booklet)

a. Solve the inequality $x - \frac{4}{x} \geq 0$ 2

b. Given the right angled triangle ABC 2



find an expression, in its simplest form, for AC.

c. The point A(4, 2) divides the interval XY internally in the ratio K : 1. If X is the point (10, -4) and Y is the point (0, 6), find the value of K, in its simplest form. 2

d. The angle between the lines
 $y = mx + 3$ and $x - 3y + 9 = 0$ is 45° . 2

Find the exact value(s) of m .

e. Simplify completely the expression

$$\frac{(n+1)! - n!}{(n-1)!} \quad 2$$

f. There are 8 people in a dinner group, 4 men and 4 women. In how many ways can they be seated around a circular table if a particular man Carl refuses to sit next to a certain woman Amanda. 2

END OF QUESTION 6

Question 7: (12 marks) (Use a SEPARATE writing booklet)

- a. If each distinct arrangement of the letters of

1

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is called a word, then how many words are possible?

- b. Find the number of ways in which the 7 letters of the word PYRAMID can be arranged in a row.

- i. without restriction

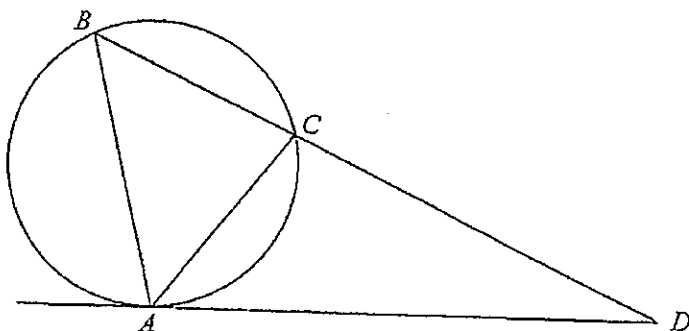
1

- ii. so that the 5 consonants are in alphabetical order from left to right.

1

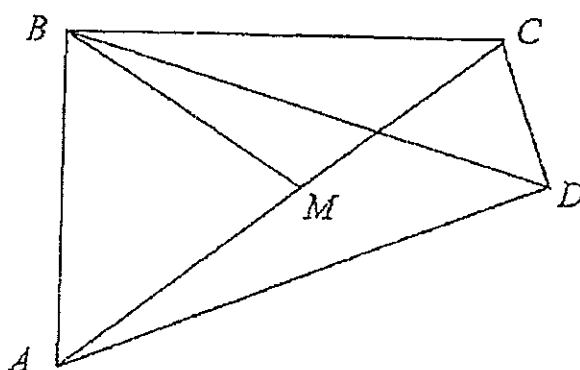
- c.

2



In the diagram, ABC is a triangle inscribed in a circle with $BC = AC$. The tangent to the circle at A meets BC produced at D . Show that AC bisects $\angle DAB$.

- d.



In the diagram, $ABCD$ is a quadrilateral in which $\angle ABC = \angle ADC = 90^\circ$. M is the midpoint of AC .

- i. Copy the diagram and explain why $ABCD$ is a cyclic quadrilateral.

1

- ii. Show that $\angle BMC = 2\angle BDC$

2

(Question 7 continues over the page)

Question 7 continued:

- e. i. Use the substitution $t = \tan \frac{x}{2}$
- to show that $\sec x + \tan x = \frac{1+t}{1-t}$ 2
- ii. Hence, solve the equation 2
- $\sec x + \tan x = -1$ for $0^\circ \leq x^\circ \leq 360^\circ$

END OF QUESTION 7

Question 8: (12 marks) (Use a SEPARATE writing booklet)

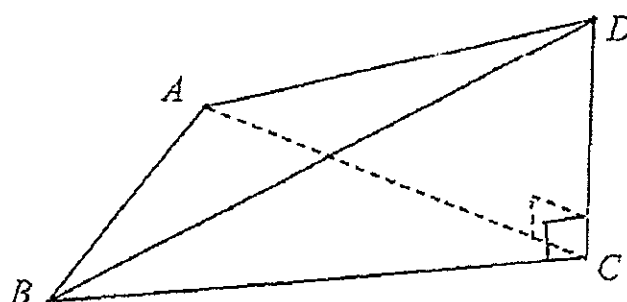
- a. Find the number of ways a committee can be chosen from 6 students, 5 parents and 4 teachers.

- i. without restriction. 1
- ii. so that the committee contains a majority of students. 2

- b. Solve the equation: 2

$$2\cos^2 x = -3 \sin x \text{ for } 0^\circ \leq x^\circ \leq 360^\circ$$

c.



In the diagram a vertical tower CD of height h metres stands with its bottom C on horizontal ground. A and B are two points on the ground such that $AB = 28$ metres.

From B the bearing of the tower is 050° and the angle of elevation of the top D of the tower is 30° . From A the bearing of the tower is 110° and the angle of elevation of the top D of the tower is 60° . Find the exact height of the tower.

- i. By first drawing a "bird's eye view" diagram indicating $\triangle ABC$ show that $\angle ACB = 60^\circ$. 1
- ii. Hence, find the exact height of the tower. 3

- d. i. Express $\tan(A + B)$ in terms of $\tan A$ and $\tan B$. 1
- ii. Hence, show that, if A , B and C are the vertex angles of a triangle then, 2
- $$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

END OF QUESTION 8

Question 9: (12 marks) (Use a SEPARATE writing booklet)

a. Factorise fully $54x^3 + 16$

1

b. i. Show that

$$\frac{1 + \tan x}{1 - \tan x} - \frac{1 - \tan x}{1 + \tan x} = 2 \tan 2x$$

2

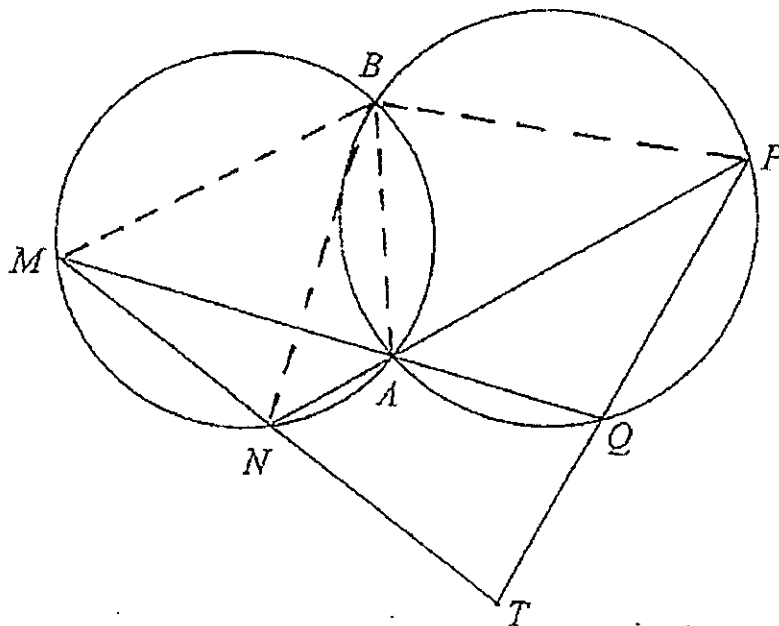
ii. Hence, find the value of:

$$\frac{1 + \tan 75^\circ}{1 - \tan 75^\circ} - \frac{1 - \tan 75^\circ}{1 + \tan 75^\circ}$$

1

in simplest exact form.

c.



3

In the diagram, two circles intersect at A and B . Points M and N lie on one circle and points P and Q lie on the other circle such that MAQ and NAP are straight lines. MN produced and PQ produced meet at T .

(Also, lines MB , NB , BA and BP are constructed for you.)

Show that $TNBP$ is a cyclic quadrilateral.

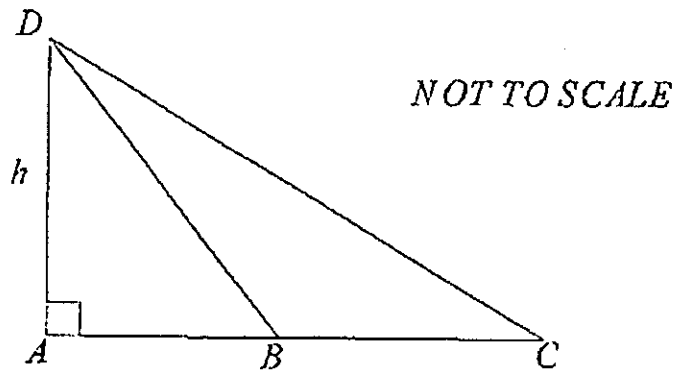
(Question 9 continues over the page)

Question 9 continued:

d.

(i) Prove that $\sin(45^\circ + \theta) \sin(45^\circ - \theta) = \frac{1}{2} \cos 2\theta$ 2

(ii) In the diagram, D is the top of a tower h metres high. The angles of depression of the points B and C are $(45^\circ + \theta)$ and $(45^\circ - \theta)$ respectively 3



Using the result in (i) show that $BC = 2h \tan 2\theta$

END OF EXAMINATION

Start here.

SECTION 1

$$\frac{9^n - 6^n}{9^n - 4^n} = \frac{3^n(3^n - 2^n)}{(3^n - 2^n)(3^n + 2^n)} = \frac{3^n}{3^n + 2^n} \quad (D)$$

$$\begin{aligned} 2. R \cos(x + \alpha) &= R(\cos x \cos \alpha - \sin x \sin \alpha) \\ &= R \cos x \cos \alpha - R \sin x \sin \alpha \end{aligned}$$

$$\begin{aligned} \therefore R \cos \alpha &= -1 \\ R \sin \alpha &= -1 \end{aligned} \quad (A)$$

$$\frac{{}^nP_3}{{}^nC_3} = \frac{\frac{n!}{(n-3)!3!}}{\frac{n!}{(n-3)!}} = \frac{1}{3!}$$

$$= \frac{1}{6} \quad (D)$$

$$\begin{aligned} 4. 7\sqrt{x^5} - \sqrt{4x} + 3\sqrt{x^6} &= 7\sqrt{x^4 \cdot x} - 2\sqrt{x} + 3x^3 \\ &= 7x^2\sqrt{x} - 2\sqrt{x} + 3x^3 \\ &= 3x^3 + (7x^2 - 2)\sqrt{x} \end{aligned} \quad (B)$$

$$\therefore a = 3x^3 \quad b = 7x^2 - 2$$

$$5. AT^2 = BT \times CT$$

$$10^2 = BT \times 8$$

$$100 = 8 \times BT$$

$$12.5 = BT \quad (C)$$

$$\begin{aligned} \therefore BC &= 12.5 - 8 \\ &= 4.5 \text{ cm.} \end{aligned}$$

Answers.

1. D

2. A

3. D

4. B

5. C

SECTION 11

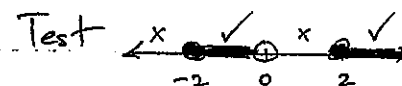
$$Q6. a) x - \frac{4}{x} \geq 0$$

$$\text{Consider } x - \frac{4}{x} = 0 \quad x \neq 0$$

$$x^2 - 4 = 0$$

$$x^2 = 4$$

$$x = \pm 2$$



$$-2 \leq x < 0 \text{ or } x \geq 2$$

b) Using Pythagoras

$$(2xy)^2 + (x^2 - y^2)^2 = AC^2$$

$$4x^2y^2 + x^4 - 2x^2y^2 + y^4 = AC^2$$

$$x^4 + 2x^2y^2 + y^4 = AC^2$$

$$(x^2 + y^2)^2 = AC^2$$

$$\therefore AC = x^2 + y^2$$

(positive since a length)

$$c) (10, -4) \quad (0, 6)$$

$$k:1$$

↓

$$(4, 2)$$

using the x co-ord:

$$1(10) + k(0) = 4$$

$$k+1$$

$$10 = 4k + 4$$

$$6 = 4k$$

$$\frac{6}{4} = k$$

$$\therefore k = \frac{3}{2}$$

$$(d) \text{ Using } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{m - \frac{1}{3}}{1 + \frac{1}{3}m} \right|$$

$$1 = \left| \frac{3m - 1}{3 + m} \right|$$

$$1 = \left| \frac{3m - 1}{3 + m} \right|$$

$$\therefore \frac{3m - 1}{3 + m} = 1 \text{ or } \frac{3m - 1}{3 + m} = -1$$

$$3m - 1 = 3 + m \text{ or } 3m - 1 = -3 - m$$

$$2m = 4$$

$$4m = -2$$

$$m = 2$$

$$m = -\frac{1}{2}$$

$$\therefore m = 2 \text{ or } -\frac{1}{2}$$

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$$(e) \frac{(n+1)! - n!}{(n-1)!}$$

$$= \frac{n!(n+1-1)}{(n-1)!}$$

$$= \frac{n!(n)}{(n-1)!}$$

$$= n \times n$$

$$= n^2$$

(f) Total seatings = $7!$

Ways together = $6! \times 2$

$$\therefore \text{Ways apart} = 7! - 6! \times 2$$

$$= 3600$$

Q7 a) Arrangements = $\frac{7!}{3!2!}$

$$= 420$$

b) (i) $7! = 5040$

(ii) Ways the consonants can be arranged = $5!$

(alphabetical order is only 1 of these!)

You may ask for an extra Writing Booklet if you need more space.

Ways within the 7 letter arrangement

$$= \frac{7!}{5!}$$

$$= 42$$

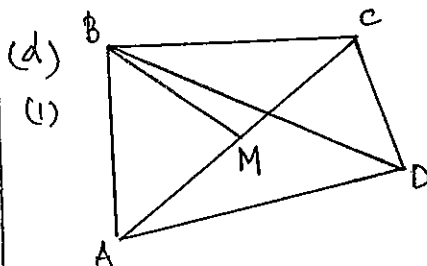
(c) let $\hat{CAD} = \theta$

$\therefore \hat{ABC} = \theta$ (alternate segment theorem)

also $\hat{BAC} = \theta$ (equal base $\angle$ s in isosceles $\triangle$)

$\therefore \hat{ABC} = \hat{BAC} = \theta$

$\therefore AC$ bisects $\hat{DAB}$



ABCD is a cyclic quadrilateral since $\hat{ABC} = \hat{ADC} = 90^\circ$ (given data)

and opposite angles in a cyclic quadrilateral are equal

Start here.

(ii) Since $\hat{ABC} = 90^\circ$

AC must be a diameter (angle in a semicircle = 90°)

$\therefore M$ is the centre of the circle (midpoint of diameter)

$$\therefore \hat{BMC} = 2 \times \hat{BDC}$$

(angle at the centre is twice the angle at the circumference standing on the same arc)

(e)(i) $\cos x = \frac{1-t^2}{1+t^2} \therefore \sec x = \frac{1+t^2}{1-t^2}$

$$\tan x = \frac{2t}{1-t^2}$$

$$\therefore \sec x + \tan x = \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2}$$

$$= \frac{1+2t+t^2}{1-t^2}$$

$$= \frac{(1+t)^2}{(1-t)(1+t)}$$

$$= \frac{1+t}{1-t}$$

as req.

(ii.) $\therefore \frac{1+t}{1-t} = -1$

$$1+t = -1+t$$

$$0 = -2$$

$\therefore$ no solutions

but we must test $x = 180^\circ$

LHS =

$$\sec 180^\circ + \tan 180^\circ$$

$$= -1 + 0$$

$$= -1$$

$\therefore 180^\circ$ is a solution

(the only solution)

Q8

(a) (i) ${}^{15}C_3 = 455$

(ii) there can be 3 students or 2.

$$\therefore {}^6C_3 + {}^6C_2 \times {}^9C_1$$

$$= 20 + 15 \times 9$$

$$= 155$$

(b) $2\cos^2 x = -3\sin x$

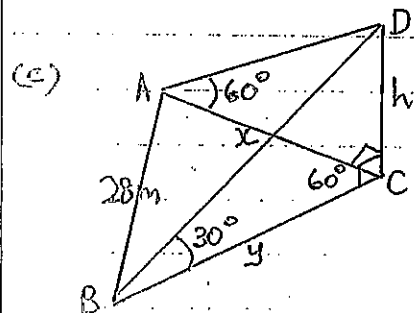
$$2(1-\sin^2 x) = -3\sin x$$

$$0 = 2\sin^2 x + 3\sin x - 2$$

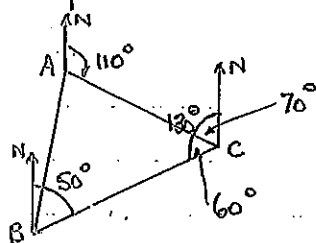
$$0 = (2\sin x + 1)(\sin x - 2)$$

$$\therefore \sin x = -\frac{1}{2} \text{ or } \sin x = -2 \text{ (no solns)}$$

$$\therefore x = 210^\circ, 330^\circ$$



This diagram would also help.



From the top diagram

$$\tan 60^\circ = \frac{h}{x}$$

$$x = \frac{h}{\tan 30^\circ}$$

$$x = \frac{h}{\sqrt{3}}$$

$$\text{Similarly } y = \sqrt{3}h$$

Using the cosine rule

$$28^2 = x^2 + y^2 - 2xy \cos 60^\circ$$

$$784 = \frac{h^2}{3} + 3h^2 - 2h^2 \times \frac{1}{2}$$

$$784 = \frac{10h^2}{3} - h^2$$

$$784 = \frac{7h^2}{3}$$

$$2352 = 7h^2$$

$$336 = h^2$$

$$\sqrt{336} = h$$

$$\therefore h = 4\sqrt{21} \text{ m}$$

$$(d) (i) \tan(A+B)$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

(ii) if A, B, C are the 3 angles of a Δ then $A+B+C = 180^\circ$

$$\therefore A+B = 180^\circ - C$$

$$\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(180^\circ - C)$$

$$\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\tan A + \tan B = -\tan C (1 - \tan A \tan B)$$

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$$\therefore \tan A + \tan B = -\tan C + \tan A \tan B \tan C$$

$$\therefore \tan A + \tan B + \tan C = \tan A \tan B \tan C \quad (c) \hat{MAB} = \hat{MNB}$$

as req.

($\angle$'s at the circumference standing on the same arc are equal)

Q9

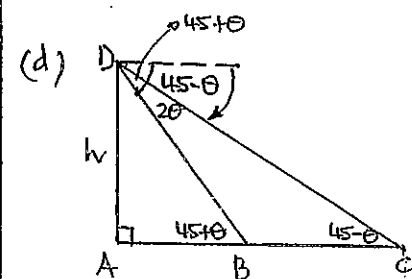
$$\begin{aligned} (a) \quad & 54x^3 + 16 \\ &= 2(27x^3 + 8) \\ &= 2(3x+2)(9x^2 - 6x + 4) \end{aligned}$$

$\hat{MAB} = \hat{BPQ}$ (exterior $\angle$ of cyclic quadrilateral is equal to interior opposite $\angle$)

and $\therefore \hat{MNB} = \hat{BPQ}$

$$\begin{aligned} (b) \text{ LHS} &= \frac{(1+\tan x)^2 - (1-\tan x)^2}{(1-\tan x)(1+\tan x)} \text{ and } \therefore \text{TNBP is a cyclic quadrilateral} \\ &= \frac{1+2\tan x + \tan^2 x - 1+2\tan x - \tan^2 x}{(1-\tan x)(1+\tan x)} \text{ since exterior } \angle \text{ of cyclic quadrilateral} = \text{interior opposite angle.} \\ &= \frac{4\tan x}{1-\tan^2 x} \end{aligned}$$

$$\begin{aligned} &= 2 \times \frac{2\tan x}{1-\tan^2 x} \\ &= 2 \times \tan 2x \\ &= 2 \tan 2x \text{ as req.} \end{aligned}$$



$$(ii) \therefore x = 75^\circ$$

$$\begin{aligned} \text{LHS} &= 2 \tan 150^\circ \\ &= 2 \times -\frac{1}{\sqrt{3}} \end{aligned}$$

$$\begin{aligned} &= -\frac{2}{\sqrt{3}} \\ &= -\frac{2\sqrt{3}}{3} \end{aligned}$$

$$\begin{aligned} (1) \quad & \sin(45+\theta) \sin(45-\theta) \\ &= (\sin 45^\circ \cos \theta + \sin \theta \cos 45^\circ) \\ &\quad \times (\sin 45^\circ \cos \theta - \sin \theta \cos 45^\circ) \\ &= \sin^2 45^\circ \cos^2 \theta - \sin^2 \theta \cos^2 45^\circ \\ &= \frac{1}{2} \cdot \cos^2 \theta - \frac{1}{2} \sin^2 \theta \end{aligned}$$

You may ask for an extra Writing Booklet if you need more space.

Start here.

$$= \frac{1}{2} (\cos^2 \theta - \sin^2 \theta)$$

$$= \frac{1}{2} \cos 2\theta \text{ as req.}$$

(ii) In $\triangle ADB$

$$\sin(45+\theta) = \frac{h}{DB} \quad \text{--- (1)}$$

In $\triangle BDC$, using sine rule

$$\frac{BC}{\sin 2\theta} = \frac{DB}{\sin(45-\theta)} \quad \text{--- (2)}$$

From (1)

$$DB = \frac{h}{\sin(45+\theta)}$$

$$\text{In (2)} \quad \frac{BC}{\sin 2\theta} = \frac{h}{\sin(45+\theta) \sin(45-\theta)}$$

$$\frac{BC}{\sin 2\theta} = \frac{h}{\frac{1}{2} \cos 2\theta}$$

$$BC = \frac{2h \sin 2\theta}{\cos 2\theta}$$

$$= 2h \tan 2\theta$$

as req.